

MULTIHALUDET: Multilingual Hallucination Detection via LLM Hidden State Probing

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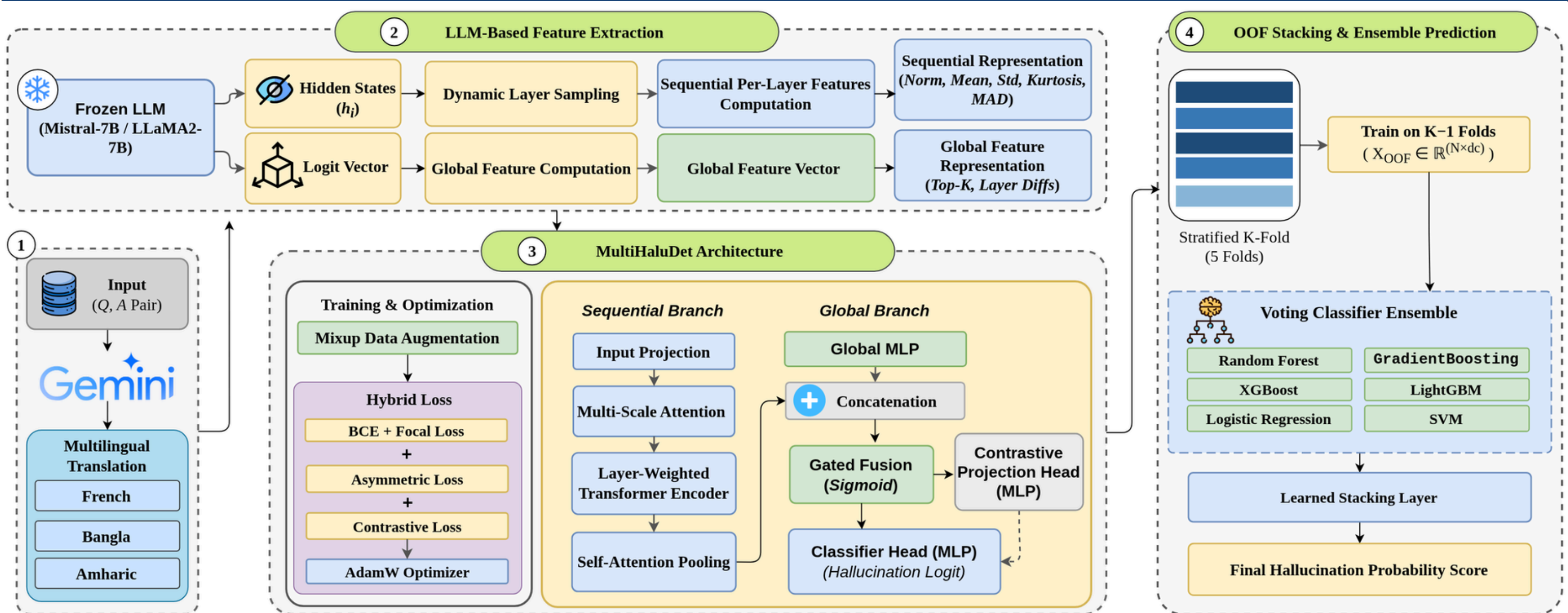
Motivation

Large Language Models (LLMs) frequently generate hallucinations, limiting their reliable use in real-world applications. Existing hallucination detectors either depend on external knowledge retrieval, require multiple inference passes, or rely on hidden representations from only a single token or a few transformer layers. These approaches often miss factual inconsistency signals that are distributed across the model's internal computation, particularly in multilingual and resource-constrained languages. This motivates a framework that analyzes the evolution of hidden representations throughout the entire transformer depth to enable more robust and language-agnostic hallucination detection.

Contributions

- We introduce MULTIHALUDET, a four-stage framework that probes hidden-state trajectories of frozen LLMs without requiring language-specific fine-tuning.
- We evaluate the framework on HaluEval and TriviaQA using Mistral-7B and LLaMA2-7B, observing strong detection performance (up to 98.55% AUROC) relative to the baselines considered in this work.
- We evaluate the framework's cross-lingual generalization across high-resource, medium-resource, and low-resource languages; and observe that detection signals are retained to varying degrees across these settings without language-specific fine-tuning.

Proposed Framework



Main Results

Method	HaluEval		TriviaQA	
	Mistral-7B	LLaMA2-7B	Mistral-7B	LLaMA2-7B
P(True)	49.7	46.7	48.3	42.3
AvgProb	43.6	42.1	48.5	44.1
AvgEnt	49.7	47.3	47.6	41.1
EUBHD	70.5	71.9	80.6	80.5
Unigram	62.3	58.2	59.5	56.8
NLI	63.1	61.3	61.4	59.4
INSIDE	76.0	74.5	81.3	81.7
SAPLMA	89.4	87.0	84.1	80.0
MIND	94.5	86.1	84.5	79.4
Probe@Exact	93.4	88.3	84.1	81.3
Neural ODEs	91.2	89.5	83.7	81.7
Neural CDEs	95.4	91.4	84.1	83.7
Neural SDEs	93.7	92.8	85.1	81.0
MultiHaluDet	98.43	98.55	98.30	98.26

MultiHaluDet consistently achieves state-of-the-art hallucination detection across both HaluEval and TriviaQA. Unlike confidence-based methods (P(True), AvgProb, AvgEnt), which remain close to random guessing, our hidden-state trajectory modeling captures semantic inconsistencies distributed throughout the transformer depth. Compared to recent probing methods such as SAPLMA, MIND, Probe@Exact, and Neural CDEs, MultiHaluDet improves AUROC by up to 3–15%, while remaining highly stable across different backbone LLMs. These consistent gains across datasets and model architectures demonstrate the effectiveness of modeling hidden-state trajectories for robust and generalizable multilingual hallucination detection.

Cross-Lingual Results

Language	Method	HaluEval (M / D)	TriviaQA (M / D)
English (Base)	Best Baseline	95.4 / 92.8	85.1 / 83.7
	MultiHaluDet	98.4 / 98.5	98.3 / 98.2
French (High)	Best Baseline	92.1 / 89.4	81.3 / 80.5
	MultiHaluDet	96.2 / 95.8	95.5 / 94.9
Bangla (Medium)	Best Baseline	78.4 / 75.1	69.2 / 67.8
	MultiHaluDet	89.1 / 88.4	87.6 / 86.3
Amharic (Low)	Best Baseline	62.3 / 59.8	54.1 / 52.6
	MultiHaluDet	78.5 / 76.2	75.8 / 73.4

M: Mistral-7B D: LLaMA2-7B

We evaluate MultiHaluDet on four languages spanning high-resource (French), medium-resource (Bangla), and low-resource (Amharic) settings. Performance naturally decreases as language resources become scarcer, yet our framework consistently outperforms the strongest baseline across every language. This demonstrates that hallucination signals embedded in hidden-state trajectories remain largely language agnostic and transfer effectively without language-specific fine-tuning.

Discussion

MultiHaluDet demonstrates that hallucination signals emerge from the evolution of hidden representations rather than isolated layers or output confidence. Integrating sequential trajectories with global features yields robust, architecture-independent detection, enabling strong multilingual generalization across diverse language resource settings without language-specific fine-tuning.

Ablation Results

Method	Mistral-7B		LLaMA2-7B	
	HaluEval	TriviaQA	HaluEval	TriviaQA
Full	98.43	98.30	98.55	98.26
w/o MSA	91.45 (↓6.98)	90.82 (↓7.48)	92.14 (↓6.41)	91.33 (↓6.93)
w/o OOF	88.67 (↓9.76)	87.41 (↓10.89)	89.25 (↓9.30)	88.19 (↓10.07)
w/o TP	93.28 (↓5.15)	92.56 (↓5.74)	93.71 (↓4.84)	93.04 (↓5.22)

• Multi-Scale Attention captures local semantic shifts across transformer layers that are missed by standard pooling.

• Trajectory Probing models the evolution of hidden representations instead of relying on only the final layer.

• Out-of-Fold Stacking generates unbiased deep embeddings and substantially improves generalization.

• The combination of sequential and global features provides complementary hallucination signals.

Limitations & Future Work

The framework requires white-box access to hidden states and incurs additional computational overhead during feature extraction. Future work will evaluate native multilingual benchmarks, extend the approach to larger and mixture-of-experts models, and investigate trajectory-based detection for black-box LLMs.